
```
% Calculation and Verification of Frequency Deviation  
% and Modulation Index of FM Signal
```

```
clc;  
clear;  
close all;
```

Message Signal Parameters

```
Am = 2; % Message amplitude (V)  
fm = 500; % Message frequency (Hz)
```

Carrier Signal Parameters

```
Ac = 1; % Carrier amplitude (V)  
fc = 10000; % Carrier frequency (Hz)
```

FM Parameters

```
kf = 250; % Frequency sensitivity (Hz/V)  
fs = 100000; % Sampling frequency (Hz)  
T = 0.01; % Simulation duration (s)
```

Time Vector

```
t = 0:1/fs:T;
```

Message Signal

```
m = Am*cos(2*pi*fm*t);
```

Frequency Deviation

```
delta_f = kf*Am;
```

Modulation Index

```
beta = delta_f/fm;
```

FM Signal

```
fm_signal = Ac*cos(2*pi*fc*t + beta*sin(2*pi*fm*t));
```

Display Parameters

```
fprintf('Carrier Frequency (fc) = %.0f Hz\n', fc);  
fprintf('Message Frequency (fm) = %.0f Hz\n', fm);
```

```
fprintf('Message Amplitude (Am) = %.0f V\n', Am);
fprintf('Frequency Sensitivity (kf) = %.0f Hz/V\n', kf);

fprintf('\nFrequency Deviation (Delta f) = %.2f Hz\n', delta_f);
fprintf('Modulation Index (Beta) = %.2f\n', beta);
```

```
Carrier Frequency (fc) = 10000 Hz
Message Frequency (fm) = 500 Hz
Message Amplitude (Am) = 2 V
Frequency Sensitivity (kf) = 250 Hz/V
```

```
Frequency Deviation (Delta f) = 500.00 Hz
Modulation Index (Beta) = 1.00
```

Verification

```
beta_verify = delta_f/fm;

fprintf('Verified Modulation Index = %.2f\n', beta_verify);

Verified Modulation Index = 1.00
```

Verification Check

```
tolerance = 1e-6;

if abs(beta - beta_verify) < tolerance
    fprintf('\nVerification Successful\n');
else
    fprintf('\nVerification Failed\n');
end
```

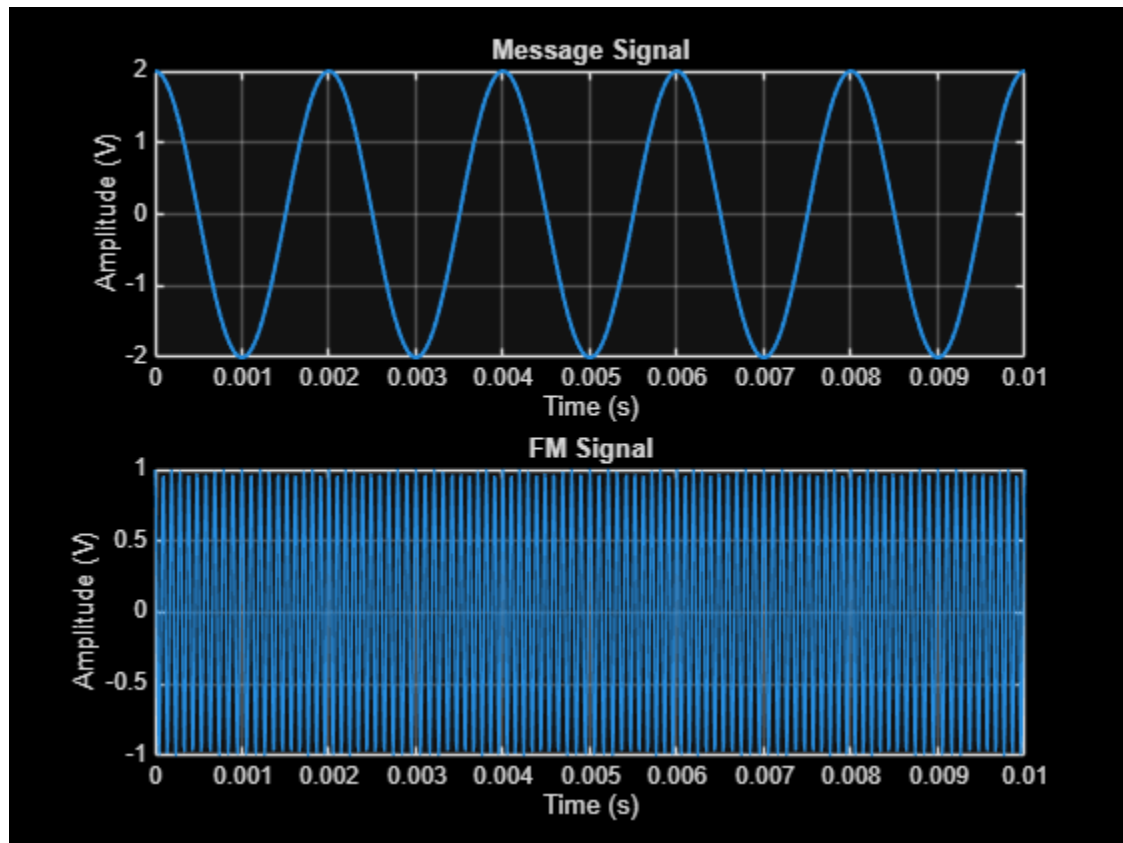
```
Verification Successful
```

Plotting

```
figure('Name', 'FM Frequency Deviation and Modulation Index', ...
       'NumberTitle', 'off');

subplot(2,1,1);
plot(t,m, 'LineWidth', 1.5);
grid on;
title('Message Signal');
xlabel('Time (s)');
ylabel('Amplitude (V)');

subplot(2,1,2);
plot(t, fm_signal, 'LineWidth', 1);
grid on;
title('FM Signal');
xlabel('Time (s)');
ylabel('Amplitude (V)');
```



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